

OC -> VM1

Сначала не тот IP указал XD

```
user@ubuntu:~$ sudo ufw delete 1
Deleting:
allow from 192.168.100.68 to any port 22 proto tcp
Proceed with operation (y|n)? y
Rule deleted
user@ubuntu:~$ sudo ufw reload
Firewall reloaded
user@ubuntu:~$ sudo ufw allow from 192.168.100.65 to any port 22 proto tcp
Rule added
user@ubuntu:~$ sudo ufw reload
Firewall reloaded
user@ubuntu:~$ sudo ufw status numbered
Status: active

      To
      --
[ 1] 22/tcp
      Action
      -----
      ALLOW IN
      From
      ----
      192.168.100.65
user@ubuntu:~$
```

VM1 -> хостовая ос

```
user@ubuntu:~$ sudo ufw allow to 192.168.100.68 port 22 proto tcp
[sudo] password for user:
Rule added
user@ubuntu:~$ sudo ufw status
Status: active

To
--
22/tcp
192.168.100.68 22/tcp
Action
-----
ALLOW
ALLOW
From
----
192.168.100.65
Anywhere
```

VM2 -> VM1

192.168.100.25 ← 192.168.100.28

```
user@ubuntu:~$ sudo ufw allow from 192.168.100.28 to any port 22 proto tcp
Rule added
user@ubuntu:~$ sudo ufw status
Status: active

To
--
22/tcp
192.168.100.68 22/tcp
22/tcp
Action
-----
ALLOW
ALLOW
ALLOW
From
----
192.168.100.65
Anywhere
192.168.100.28
```

192.168.100.28:any → 192.168.100.25:22

```
user@ubuntu:~$ sudo ufw allow to 192.168.100.25 port 22 proto tcp
Rule added
```

VM2 -x> хостовая ОС

```
user@ubuntu:~$ sudo ufw allow to 192.168.100.65 port 22 proto tcp
Rule added
user@ubuntu:~$ sudo ufw status
Status: active
```

To	Action	From
--	-----	----
2049	ALLOW	192.168.100.25
22/tcp	ALLOW	192.168.100.65
192.168.100.25 22/tcp	ALLOW	Anywhere
192.168.100.65 22/tcp	ALLOW	Anywhere

Еще добавил ssh-ключ на сервер

Генирируем ключ

```
user@ubuntu:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/user/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/user/.ssh/id_ed25519
Your public key has been saved in /home/user/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:m8TmXrS1aafBk5pJ4XIMtfRl9Z5Bduw4TNXAKl86Nto user@ubuntu
```

Добавляем на сервер

```
user@ubuntu:~$ ssh-copy-id -i ~/.ssh/id_ed25519.pub user@192.168.100.28
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/user/
id_ed25519.pub"
The authenticity of host '192.168.100.28 (192.168.100.28)' can't be established.
ED25519 key fingerprint is SHA256:o1fMMKBByNASv/mDYF8lnBYlWgBBhXPN80mPQa5P.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to find
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are
interested now it is to install the new keys
user@192.168.100.28's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'user@192.168.100.28'"
and check to make sure that only the key(s) you wanted were added.

user@ubuntu:~$
```

Коннектимся

```
id_ed25519      known_hosts
user@ubuntu:~$ ssh-add ~/.ssh/id_ed25519
Enter passphrase for /home/user/.ssh/id_ed25519:
Identity added: /home/user/.ssh/id_ed25519 (user@ubuntu)
user@ubuntu:~$ ssh user@192.168.100.28
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Apr 22 06:33:52 PM UTC 2026

System load:      0.0                Processes:        148
Usage of /home:   0.0% of 973MB      Users logged in: 1
Memory usage:     6%                IPv4 address for enp0s3: 192.168.100.28
Swap usage:       0%

⇒ / is using 99.9% of 11.21GB
```

Теперь задание от Тайлера Дёрдена

Прошел туториал по пакеру на официальном сайте. Прохождение в отдельном PDF.

Для начала надо поставить плагин для VirtualBox.

```
packer {  
  required_plugins {  
    virtualbox = {  
      version = "~> 1"  
      source  = "github.com/hashicorp/virtualbox"  
    }  
  }  
}
```

Плагин качает is

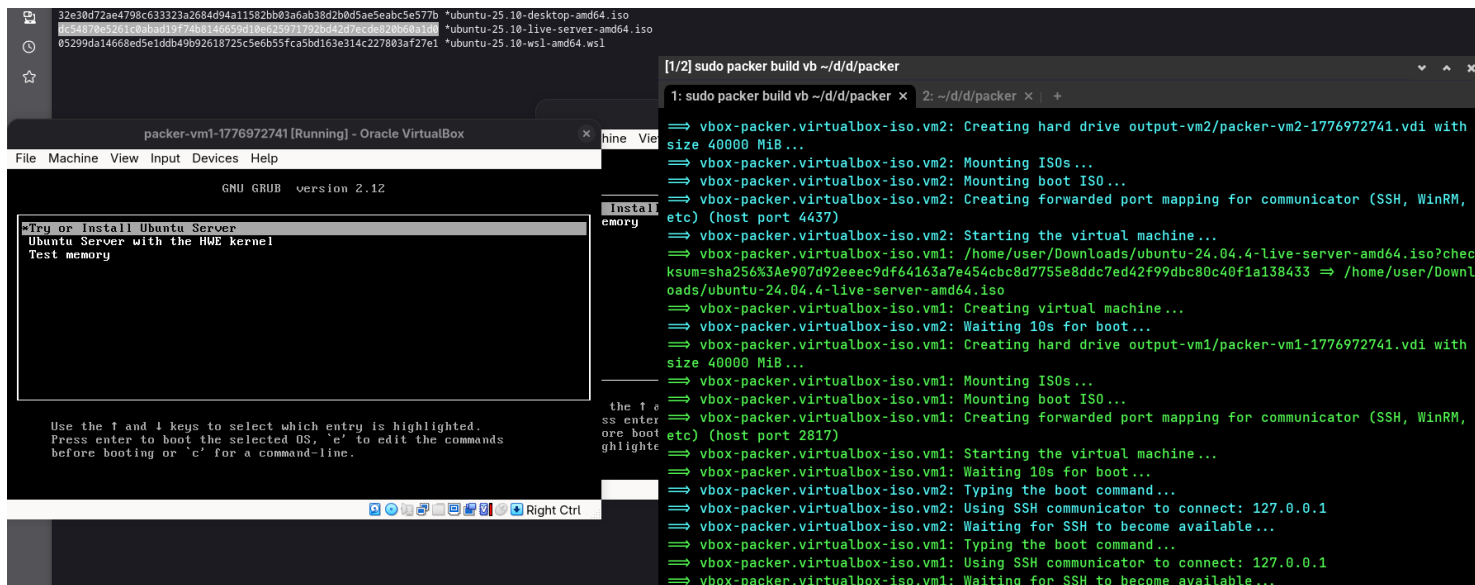
Есть несколько вариантов создания image.

Builders

- [virtualbox-iso](#) - Starts from an ISO file, creates a brand new VirtualBox VM, installs an OS, provisions software within the OS, then exports that machine to create an image. This is best for people who want to start from scratch.
- [virtualbox-ovf](#) - This builder imports an existing OVF/OVA file, runs **provisioners** on top of that VM, and exports that machine to create an image. This is best if you have an existing VirtualBox VM export you want to use as the source. As an additional benefit, you can feed the artifact of this builder back into itself to iterate on a machine.
- [virtualbox-vm](#) - This builder uses an existing VM to run defined provisioners on top of that VM, and optionally creates a snapshot to save the changes applied from the provisioners. In addition the builder is able to export that machine to create an image. The builder is able to attach to a defined snapshot as a starting point, which could be defined statically or dynamically via a variable.

Буду использовать builder virtualbox-iso.

После настройки темплейта и его запуска возникла проблема, что операционная система не устанавливается автоматически XD.



Босс качалки, он же unattended install операционной системы.

Конфиг самой операционной системы составил. Он делается на основе документации по autoinstall на сайте ubuntu.

Теперь еще одна проблема с недоступностью локалхоста по NAT, потому что оказывается с 7 версии VirtualBox сеть не доступна с локалхоста и операционная система не может достучаться до веб-сервера пакера. Для начала надо в блок source добавить настройку VirtualBox, чтобы он разрешил доступ к локалхосту через NAT сеть.

```
vboxmanage = [{"modifyvm", "{{.Name}}", "--nat-localhostreachable1", "on"}]
```

Короче нифига не помогло, пробую через CD.

```
jsnonschema.validate(ai_data, self.autoinstall_schema)
File "/snap/subiquity/7020/usr/lib/python3/dist-packages/jsnonschema/validators.py", line 934, in validate
raise error
jsnonschema.exceptions.ValidationError: Additional properties are not allowed ('groups' was unexpected)

Failed validating 'additionalProperties' in schema:
{'additionalProperties': False,
 'properties': {'hostname': {'type': 'string'},
 'password': {'type': 'string'},
 'realname': {'type': 'string'},
 'username': {'type': 'string'}},
 'required': ['username', 'hostname', 'password'],
 'type': 'object'}

On instance:
{'groups': ['sudo'],
 'hostname': 'ubu-packer',
 'password': '$6$aQ_DH/tCpwAETs8HxH0THtrnBXHe3J.4rtyJ7g30Pp8CpPAWRICF20m3wsz43DNms4af620KurmBavcMa0YWEpm1l/xnKEVHds02G/',
 'realname': 'Packer Installer',
 'username': 'installer'}

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
File "/snap/subiquity/7020/lib/python3.10/site-packages/subiquity/server/server.py", line 1000, in start
self.load_autoinstall_config(only_early=False)
File "/snap/subiquity/7020/lib/python3.10/site-packages/subiquitycore/context.py", line 160, in decorated_sync
return meth(self, **kw)
File "/snap/subiquity/7020/lib/python3.10/site-packages/subiquity/server/server.py", line 739, in load_autoinstall_config
controller.setup_autoinstall()
File "/snap/subiquity/7020/lib/python3.10/site-packages/subiquity/server/controller.py", line 90, in setup_autoinstall
self.validate_autoinstall(ai_data)
File "/snap/subiquity/7020/lib/python3.10/site-packages/subiquity/server/controller.py", line 70, in validate_autoinstall
raise new_exception from original_exception
subiquity.server.autoinstall.AutoinstallValidationError: Malformed autoinstall in 'identity' section
2026-04-25 21:44:23.259 INFO root:38 finish: subiquity/Meta/status_GET: SUCCESS: 200 {'state': "ERROR", "confirming_tty": "",
nron": null, "nonreportable_error"...
root@ubuntu-server:/var/log/installer#
```

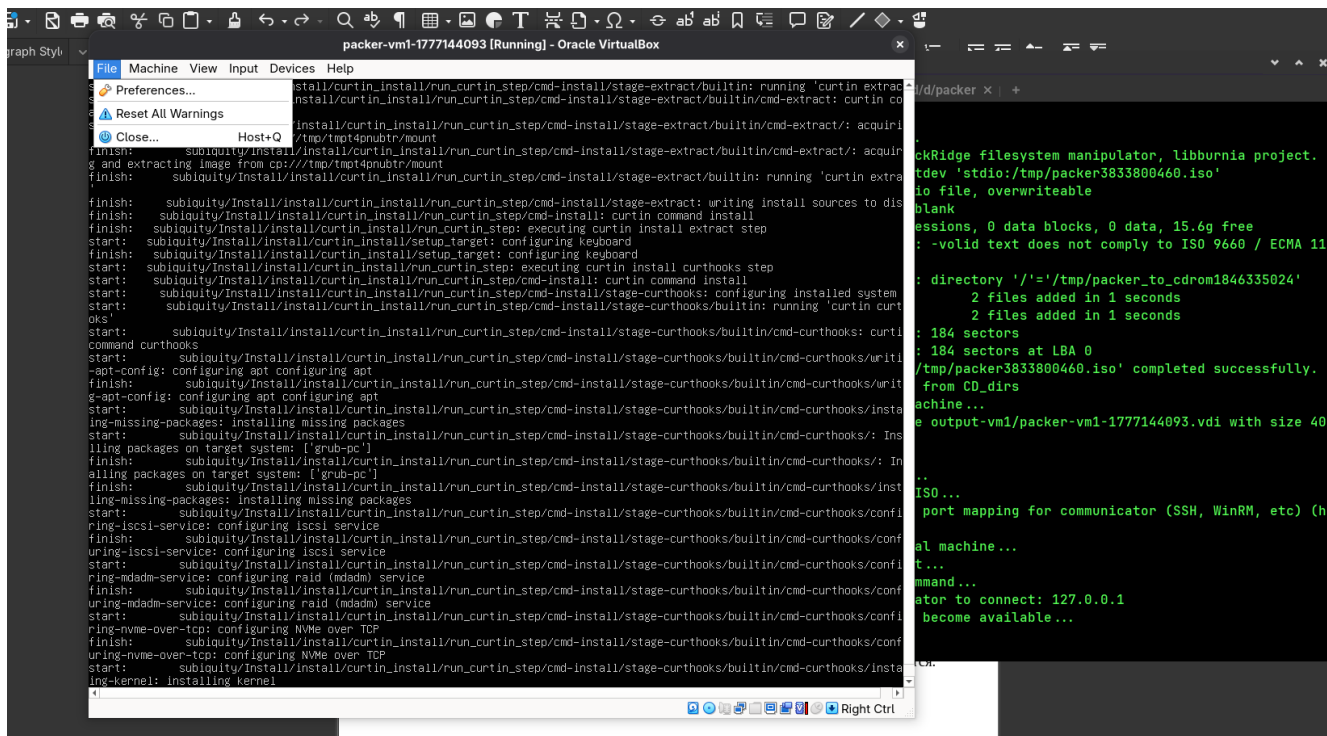
```
current: std
a status : is
summary: 0 s
iso : WARNING

to ISO image
iso : UPDATE :
iso : UPDATE :
image produced
ten to medium
ing to 'stdio:
copying paths
ing virtual m
ing hard driv

ing ISOs...
ing cd_files
ing boot ISO.
ing forwarded

ing the virtu
ing 5s for boo
ing the boot co
ing SSH communic
ing for SSH to
```

Пишет, что groups не должно быть в identity..... В документации оно допускается. Заменял identity на user-data и заработало.



```
packer-vm1-1777144093 [Running] - Oracle VirtualBox

File Machine View Input Devices Help
Preferences...
Reset All Warnings
Close... Host+Q

install/curtin_install/run_curtin_step/cmd-install/stage-extract/builtin: running 'curtin extra
install/curtin_install/run_curtin_step/cmd-install/stage-extract/builtin/cmd-extract: curtin co
install/curtin_install/run_curtin_step/cmd-install/stage-extract/builtin/cmd-extract/: acquiri
/tmp/tmp4qnbvtr/mount
install/curtin_install/run_curtin_step/cmd-install/stage-extract/builtin/cmd-extract/: acquiri
g and extracting image from cp:///tmp/tmp4qnbvtr/mount
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-extract/builtin: running 'curtin extra
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-extract: writing install sources to dis
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install: curtin command install
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks: configuring installed system.
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin: running 'curtin curt
oks'
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks: curtin
command curtin_hooks
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/writi
g-apt-config: configuring apt configuring apt
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/writi
g-apt-config: configuring apt configuring apt
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/insta
lling-missing-packages: installing missing packages
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/: Ins
talling packages on target system: ['grub-pc']
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/: Ins
talling packages on target system: ['grub-pc']
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/insta
lling-missing-packages: installing missing packages
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/confi
ring-iscsi-service: configuring iscsi service
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/confi
ring-iscsi-service: configuring iscsi service
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/confi
ring-mdadm-service: configuring raid (mdadm) service
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/confi
ring-mdadm-service: configuring raid (mdadm) service
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/confi
ring-nvme-over-tcp: configuring NVMe over TCP
finish: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/confi
ring-nvme-over-tcp: configuring NVMe over TCP
start: subiquity/Install/Install/curtin_install/run_curtin_step/cmd-install/stage-curtin_hooks/builtin/cmd-curtin_hooks/insta
lling-kernel: installing kernel

d/packer x | +
ackRidge filesystem manipulator, libburnia project.
tdev 'stdio:/tmp/packer383800460.iso'
io file, overwriteable
blank
essions, 0 data blocks, 0 data, 15.6g free
: -valid text does not comply to ISO 9660 / ECMA 11
: directory '/!=/tmp/packer_to_cdrom1846335024'
2 files added in 1 seconds
2 files added in 1 seconds
: 184 sectors
: 184 sectors at LBA 0
/tmp/packer383800460.iso' completed successfully.
from CD_dirs
achine...
e output-vm1/packer-vm1-1777144093.vdi with size 40
..
ISO...
port mapping for communicator (SSH, WinRM, etc) (h
al machine...
t...
command...
ator to connect: 127.0.0.1
become available...
```

```

=> vbox-packer.virtualbox-iso.vm1: Executing custom VBoxManage commands...
=> vbox-packer.virtualbox-iso.vm1: Executing: modifyvm packer-vm1-1777198068 --nat-localhostreachable1 on
=> vbox-packer.virtualbox-iso.vm1: Starting the virtual machine...
=> vbox-packer.virtualbox-iso.vm1: Waiting 5s for boot...
=> vbox-packer.virtualbox-iso.vm1: Typing the boot command...
=> vbox-packer.virtualbox-iso.vm1: Using SSH communicator to connect: 127.0.0.1
=> vbox-packer.virtualbox-iso.vm1: Waiting for SSH to become available...
=> vbox-packer.virtualbox-iso.vm1: Connected to SSH!
=> vbox-packer.virtualbox-iso.vm1: Uploading VirtualBox version info (7.2.8)
=> vbox-packer.virtualbox-iso.vm1: Uploading VirtualBox guest additions ISO...
=> vbox-packer.virtualbox-iso.vm1: Provisioning with shell script: /tmp/packer-shell3023365192
=> vbox-packer.virtualbox-iso.vm1: ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMGSwLXvWcPv9J/PN9nUPvapyK31T7ShAWNFM
@archPasswordAuthentication noRules updated
=> vbox-packer.virtualbox-iso.vm1: Rules updated (v6)
=> vbox-packer.virtualbox-iso.vm1: Command may disrupt existing ssh connections. Proceed with operation (y|n)
=> vbox-packer.virtualbox-iso.vm1: Gracefully halting virtual machine...
=> vbox-packer.virtualbox-iso.vm1: Preparing to export machine...
=> vbox-packer.virtualbox-iso.vm1: Deleting forwarded port mapping for the communicator (SSH, WinRM, etc) (h
9)
=> vbox-packer.virtualbox-iso.vm1: Exporting virtual machine...
=> vbox-packer.virtualbox-iso.vm1: Executing: export packer-vm1-1777198068 --output output-vm1/packer-vm1-17
=> vbox-packer.virtualbox-iso.vm1: Cleaning up floppy disk...
=> vbox-packer.virtualbox-iso.vm1: Deregistering and deleting VM...
Build 'vbox-packer.virtualbox-iso.vm1' finished after 4 minutes 47 seconds.

```

Запускаю VM и проверяю доступ по SSH с помощью сертификата.

```

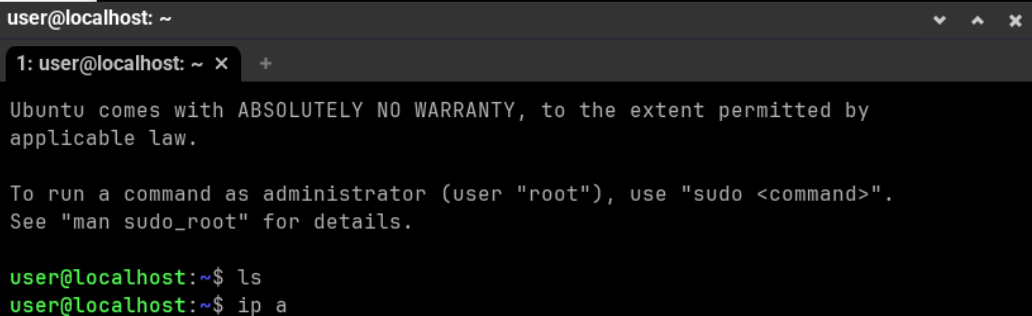
localhost login: installer
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sun Apr 26 01:19:44 PM UTC 2026

System load: 0.0               Processes:                    112
Usage of /:  12.5% of 38.15GB   Users logged in:             0
Memory usage: 11%              IPv4 address for enp0s3: 192.168.100.29
Swap usage:  0%

```



The image shows a terminal window with system information and a terminal emulator window below it.

Terminal window content:

```

user@localhost: ~

1: user@localhost: ~ x +

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

user@localhost:~$ ls
user@localhost:~$ ip a

```

Terminal emulator window content:

```

user@localhost: ~

1: user@localhost: ~ x +

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

user@localhost:~$ ls
user@localhost:~$ ip a

```

Всё работает.